

A Translation-Generator Obstruction to the Proposed Mehler–Fock H^∞ Extension

Literature-implied answer for arXiv:1509.00133

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Status. Literature-implied answer: a full negative answer to automatic extension in the standard, holomorphic-functional-calculus sense. This is not claimed as a new theorem.

Source question

Blower and Doust interpret their hypergroup representation as a functional calculus

$$\Phi_A : \mathcal{A} \longrightarrow \mathcal{L}(E), \quad \Phi_A(\psi) = \psi(A),$$

where $\mathcal{A}$ is the range of the hypergroup Fourier transform. Remark 4.2 on printed page 11 asks whether Φ_A extends to a bounded algebra homomorphism

$$H^\infty(\Sigma_{\omega_0}) \longrightarrow \mathcal{L}(E), \quad \Sigma_\omega = \{z \in \mathbb{C} : |\operatorname{Im} z| < \omega\}.$$

Counterexample

Theorem 1. *For every $q \in (1, \infty) \setminus \{2\}$, there is a contractive strongly continuous cosine family on $E = L^q(\mathbb{R})$ to which the Mehler–Fock operational calculus of Blower–Doust applies with $\omega_0 = 1/2$, but whose map Φ_A has no bounded extension to $H^\infty(\Sigma_{1/2})$ compatible with the holomorphic functional calculus.*

Proof. Let A be the closed Fourier-multiplier operator with symbol ξ : $(Au)^\wedge(\xi) = \xi \widehat{u}(\xi)$, on its natural domain in $L^q(\mathbb{R})$. If $S_t u(x) = u(x - t)$, then

$$\cos(tA) = \frac{1}{2}(S_t + S_{-t}), \quad \|\cos(tA)\| \leq 1.$$

Consequently the hypothesis $\|\cos(tA)\| \leq \kappa \cosh(t/2)$ of Blower–Doust Proposition 6.2 holds with $\kappa = 1$. Their proposition therefore produces the Mehler–Fock hypergroup operational calculus, whose Fourier-transform range is contained in $H^\infty(\Sigma_{1/2})$.

Now let $T = \exp A$, the sectorial Fourier multiplier with symbol e^ξ . Cowling–Doust–McIntosh–Yagi, Example 5.2 and Lemma 5.3, prove that, when $q \neq 2$, this T has no bounded $H^\infty(S_\theta)$ functional calculus for any $\theta > 0$. In their example T^{iu} is translation by u , so this is exactly the logarithmic pair $A = \log T$ used above.

Suppose that the extension asked for in Remark 4.2 existed as a bounded strip functional calculus for A . Given $m \in H^\infty(S_{1/2})$, put $b(z) = m(e^z)$. The exponential maps $\Sigma_{1/2}$ into $S_{1/2}$, hence

$$b \in H^\infty(\Sigma_{1/2}), \quad \|b\|_\infty \leq \|m\|_\infty.$$

Compatibility of the holomorphic calculi gives

$$b(A) = m(e^A) = m(T).$$

The assumed bounded strip calculus would thus imply $\|m(T)\| \leq C\|m\|_\infty$ for every such m , giving T a bounded $H^\infty(S_{1/2})$ calculus. This contradicts the cited Lemma 5.3. $\square$

Why the obstruction is sharp enough

The supporting proof constructs bounded holomorphic strip functions by Gaussian interpolation of arbitrary bounded sequences on $\mathbb{Z}$. If all those functions acted boundedly on $L^q(\mathbb{R})$, de Leeuw restriction would make every bounded sequence a Fourier multiplier on $L^q(\mathbb{T})$, which is possible only for $q = 2$. Thus the failure is a genuine Banach-space multiplier obstruction, despite the contractive cosine bound.

Scope and interpretation

The theorem disproves an automatic extension statement for the hypotheses of Remark 4.2. It does not preclude positive extensions on Hilbert spaces or for operators satisfying additional square-function or multiplier estimates.

The phrase “functional calculus map” is used in its standard resolvent-compatible sense, as signaled by the source formula $\Phi_A(\psi) = \psi(A)$. A purely abstract algebra homomorphism with no compatibility or convergence axiom would not be an operator functional calculus and is outside this identification.

Literature and novelty check

The exact arXiv id and title, the quoted sentence from Remark 4.2, and searches combining “finite propagation speed”, “bounded H^∞ calculus”, and “cosine family” were checked. No later paper explicitly advertising a resolution of the remark was found. Nevertheless this result is classified as literature-implied, not new: the decisive 1996 lemma predates the source, and the source itself cites the same paper in its earlier multiplier warning.

References

- [1] G. Blower and I. Doust, *Operational calculus and integral transforms for groups with finite propagation speed*, Adv. Pure Appl. Math. 8 (2017), 265–283; arXiv:1509.00133. See Remark 4.2 and Proposition 6.2.
- [2] M. Cowling, I. Doust, A. McIntosh, and A. Yagi, *Banach space operators with a bounded H^∞ functional calculus*, J. Austral. Math. Soc. Ser. A 60 (1996), 51–89. See Example 5.2 and Lemma 5.3, printed pages 76–78.